

DBMS – PRODUCT AND CATEGORY MANAGEMENT

Project Title: Big Basket-E-commerce-Database-Management-System

1. Database Design

1.1 Category Table

Attribute	Key Type
Category_ID	Primary Key
Category Name	Unique Key
Description	-

1.2 Product Table

Attribute	Key Type
Product_ID	Primary Key
Product Name	-
Category_ID	Foreign Key
Supplier_ID	Foreign Key
Brand	-
Price	Check
Stock	Check
Expiry_date	-

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: One-to-Many (1:M) — one category contains many products

3. Table Creation

3.1 Create Category Table

```
Create table Category(  
    Category_ID int primary key,  
    Category_Name varchar2(30) unique,  
    Description varchar2(30)  
);  
Table created.
```

The Category table is created with Category_ID as the primary key and Category_name as a unique field.

3.2 Create Product Table

```
CREATE TABLE Product (  
    Product_ID INT PRIMARY KEY,  
    Product_Name VARCHAR2(50),  
    Category_ID INT REFERENCES Category(Category_ID),  
    Supplier_ID INT,  
    Brand VARCHAR2(30),  
    Price NUMBER(10,2) CHECK (Price > 0),  
    Stock INT CHECK (Stock >= 0),  
    Expiry_Date DATE  
);  
Table created.
```

The Product table is created with Product_ID as the primary key and Category_ID as a foreign key referencing the Category table, establishing a one-to-many relationship.

4. Data Insertion

4.1 Category Records

Ten category records are inserted into the Category table, followed by a commit to save the transaction.

```
INSERT INTO Category VALUES (1, 'Electronics', 'Electronic products');
1 row created.

INSERT INTO Category VALUES (2, 'Clothing', 'Clothing products');
1 row created.

INSERT INTO Category VALUES (3, 'Books', 'Educational books');
1 row created.

INSERT INTO Category VALUES (4, 'Furniture', 'Home furniture');
1 row created.

INSERT INTO Category VALUES (5, 'Groceries', 'Daily grocery items');
1 row created.

INSERT INTO Category VALUES (6, 'Footwear', 'Shoes and sandals');
1 row created.

INSERT INTO Category VALUES (7, 'Sports', 'Sports equipment');
1 row created.

INSERT INTO Category VALUES (8, 'Beauty', 'Beauty products');
1 row created.

INSERT INTO Category VALUES (9, 'Toys', 'Toys and games');
1 row created.

INSERT INTO Category VALUES (10, 'Stationery', 'Stationery items');
1 row created.
```

4.2 Product Records

Ten product records are inserted into the Product table, each linked to a category via Category_ID, followed by a commit.

```
INSERT INTO Product VALUES (101, 'Laptop', 1, 1001, 'Dell', 55000, 10, NULL);
1 row created.

INSERT INTO Product VALUES (102, 'Smartphone', 1, 1002, 'Samsung', 25000, 20, NULL);
1 row created.

INSERT INTO Product VALUES (103, 'Tablet', 1, 1003, 'Lenovo', 18000, 15, NULL);
1 row created.

INSERT INTO Product VALUES (104, 'Keyboard', 1, 1004, 'HP', 1200, 25, NULL);
1 row created.

INSERT INTO Product VALUES (105, 'Mouse', 1, 1005, 'Logitech', 800, 30, NULL);
1 row created.

INSERT INTO Product VALUES (106, 'Monitor', 1, 1006, 'LG', 15000, 12, NULL);
1 row created.

INSERT INTO Product VALUES (107, 'Printer', 1, 1007, 'Canon', 12000, 8, NULL);
1 row created.

INSERT INTO Product VALUES (108, 'Speaker', 1, 1008, 'JBL', 3500, 18, NULL);
1 row created.

INSERT INTO Product VALUES (109, 'Webcam', 1, 1009, 'Logitech', 2500, 20, NULL);
1 row created.

INSERT INTO Product VALUES (110, 'Headphones', 1, 1010, 'Sony', 4500, 22, NULL);
1 row created.
```

```
INSERT INTO Product VALUES (111, 'T-Shirt', 2, 1011, 'Puma', 1200, 30, NULL);
1 row created.

INSERT INTO Product VALUES (112, 'Jeans', 2, 1012, 'Levis', 2500, 25, NULL);
1 row created.

INSERT INTO Product VALUES (113, 'Formal Shirt', 2, 1013, 'Peter England', 1800, 20, NULL);
1 row created.

INSERT INTO Product VALUES (114, 'Kurti', 2, 1014, 'Biba', 2200, 18, NULL);
1 row created.

INSERT INTO Product VALUES (115, 'Saree', 2, 1015, 'Silk India', 4500, 12, NULL);
1 row created.

INSERT INTO Product VALUES (116, 'Hoodie', 2, 1016, 'Adidas', 2800, 15, NULL);
1 row created.

INSERT INTO Product VALUES (117, 'Jacket', 2, 1017, 'Allen Solly', 3500, 10, NULL);
1 row created.

INSERT INTO Product VALUES (118, 'Shorts', 2, 1018, 'Nike', 1500, 22, NULL);
1 row created.

INSERT INTO Product VALUES (119, 'Skirt', 2, 1019, 'Zara', 2000, 14, NULL);
1 row created.

INSERT INTO Product VALUES (120, 'Sweater', 2, 1020, 'Van Heusen', 2400, 16, NULL);
1 row created.
```

```
INSERT INTO Product VALUES (121, 'DBMS Book', 3, 1021, 'McGraw', 650, 15, NULL);
1 row created.

INSERT INTO Product VALUES (122, 'Python Programming', 3, 1022, 'OReilly', 850, 12, NULL);
1 row created.

INSERT INTO Product VALUES (123, 'Java Programming', 3, 1023, 'Pearson', 900, 10, NULL);
1 row created.

INSERT INTO Product VALUES (124, 'Data Structures', 3, 1024, 'McGraw', 750, 14, NULL);
1 row created.

INSERT INTO Product VALUES (125, 'Machine Learning', 3, 1025, 'Wiley', 1100, 8, NULL);
1 row created.

INSERT INTO Product VALUES (126, 'Artificial Intelligence', 3, 1026, 'Pearson', 1250, 9, NULL);
1 row created.

INSERT INTO Product VALUES (127, 'Web Development', 3, 1027, 'Packt', 950, 11, NULL);
1 row created.

INSERT INTO Product VALUES (128, 'Computer Networks', 3, 1028, 'McGraw', 800, 13, NULL);
1 row created.

INSERT INTO Product VALUES (129, 'Operating Systems', 3, 1029, 'Pearson', 700, 16, NULL);
1 row created.

INSERT INTO Product VALUES (130, 'Software Engineering', 3, 1030, 'Wiley', 850, 10, NULL);
1 row created.
```

```
INSERT INTO Product VALUES (131, 'Computer Table', 4, 1031, 'WoodCraft', 4500, 8, NULL);
1 row created.

INSERT INTO Product VALUES (132, 'Office Chair', 4, 1032, 'FeatherLite', 6500, 10, NULL);
1 row created.

INSERT INTO Product VALUES (133, 'Dining Table', 4, 1033, 'UrbankWood', 12000, 5, NULL);
1 row created.

INSERT INTO Product VALUES (134, 'Sofa', 4, 1034, 'IKEA', 25000, 4, NULL);
1 row created.

INSERT INTO Product VALUES (135, 'Bookshelf', 4, 1035, 'HomeTown', 7000, 7, NULL);
1 row created.

INSERT INTO Product VALUES (136, 'Bed', 4, 1036, 'Wakefit', 30000, 3, NULL);
1 row created.

INSERT INTO Product VALUES (137, 'Wardrobe', 4, 1037, 'Godrej', 22000, 6, NULL);
1 row created.

INSERT INTO Product VALUES (138, 'Study Table', 4, 1038, 'Nilkamal', 5500, 9, NULL);
1 row created.

INSERT INTO Product VALUES (139, 'Coffee Table', 4, 1039, 'UrbanLadder', 6500, 8, NULL);
1 row created.

INSERT INTO Product VALUES (140, 'TV Stand', 4, 1040, 'HomeTown', 8000, 5, NULL);
1 row created.
```

```
INSERT INTO Product VALUES (141, 'Rice', 5, 1041, 'India Gate', 800, 30, TO_DATE('30-06-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (142, 'Wheat Flour', 5, 1042, 'Aashirvaad', 450, 25, TO_DATE('15-07-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (143, 'Cooking Oil', 5, 1043, 'Fortune', 180, 40, TO_DATE('30-06-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (144, 'Sugar', 5, 1044, 'Madhur', 55, 50, TO_DATE('31-12-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (145, 'Salt', 5, 1045, 'Tata', 30, 45, TO_DATE('31-12-2028','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (146, 'Toor Dal', 5, 1046, 'Tata Sampann', 160, 35, TO_DATE('30-09-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (147, 'Biscuits', 5, 1047, 'Parle', 40, 60, TO_DATE('15-05-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (148, 'Tea', 5, 1048, 'Tata Tea', 250, 30, TO_DATE('31-08-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (149, 'Coffee', 5, 1049, 'Bru', 300, 25, TO_DATE('30-11-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (150, 'Noodles', 5, 1050, 'Maggi', 60, 50, TO_DATE('31-10-2027','DD-MM-YYYY'));
1 row created.
```

```
INSERT INTO Product VALUES (151, 'Running Shoes', 6, 1051, 'Nike', 3500, 18, NULL);
1 row created.

INSERT INTO Product VALUES (152, 'Sports Sandals', 6, 1052, 'Adidas', 2200, 15, NULL);
1 row created.

INSERT INTO Product VALUES (153, 'Formal Shoes', 6, 1053, 'Bata', 2800, 12, NULL);
1 row created.

INSERT INTO Product VALUES (154, 'Sneakers', 6, 1054, 'Puma', 3200, 20, NULL);
1 row created.

INSERT INTO Product VALUES (155, 'Slippers', 6, 1055, 'Sparx', 600, 35, NULL);
1 row created.

INSERT INTO Product VALUES (156, 'Heels', 6, 1056, 'Metro', 2500, 14, NULL);
1 row created.

INSERT INTO Product VALUES (157, 'Flip Flops', 6, 1057, 'Hawaii', 400, 40, NULL);
1 row created.

INSERT INTO Product VALUES (158, 'Loafers', 6, 1058, 'Clarks', 4500, 10, NULL);
1 row created.

INSERT INTO Product VALUES (159, 'School Shoes', 6, 1059, 'Bata', 1200, 25, NULL);
1 row created.

INSERT INTO Product VALUES (160, 'Walking Shoes', 6, 1060, 'Reebok', 2800, 18, NULL);
1 row created.
```

```
INSERT INTO Product VALUES (161, 'Football', 7, 1061, 'Nivia', 900, 25, NULL);
1 row created.

INSERT INTO Product VALUES (162, 'Cricket Bat', 7, 1062, 'SS', 2800, 12, NULL);
1 row created.

INSERT INTO Product VALUES (163, 'Cricket Ball', 7, 1063, 'SG', 450, 30, NULL);
1 row created.

INSERT INTO Product VALUES (164, 'Tennis Racket', 7, 1064, 'Yonex', 3500, 10, NULL);
1 row created.

INSERT INTO Product VALUES (165, 'Badminton Racket', 7, 1065, 'Li-Ning', 2500, 15, NULL);
1 row created.

INSERT INTO Product VALUES (166, 'Basketball', 7, 1066, 'Spalding', 1800, 14, NULL);
1 row created.

INSERT INTO Product VALUES (167, 'Volleyball', 7, 1067, 'Cosco', 1200, 18, NULL);
1 row created.

INSERT INTO Product VALUES (168, 'Skipping Rope', 7, 1068, 'Nivia', 300, 35, NULL);
1 row created.

INSERT INTO Product VALUES (169, 'Yoga Mat', 7, 1069, 'Boldfit', 800, 25, NULL);
1 row created.

INSERT INTO Product VALUES (170, 'Dumbbells', 7, 1070, 'Kobo', 1500, 12, NULL);
1 row created.
```

```

INSERT INTO Product VALUES (171, 'Face Cream', 8, 1071, 'Nivea', 450, 25, TO_DATE('31-12-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (172, 'Shampoo', 8, 1072, 'Dove', 350, 30, TO_DATE('31-12-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (173, 'Lipstick', 8, 1073, 'Lakme', 599, 20, TO_DATE('30-06-2028','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (174, 'Face Wash', 8, 1074, 'Himalaya', 250, 35, TO_DATE('31-12-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (175, 'Moisturizer', 8, 1075, 'Ponds', 400, 28, TO_DATE('30-09-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (176, 'Sunscreen', 8, 1076, 'Neutrogena', 650, 22, TO_DATE('31-12-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (177, 'Kajal', 8, 1077, 'Maybelline', 300, 25, TO_DATE('30-06-2028','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (178, 'Perfume', 8, 1078, 'Fogg', 750, 18, TO_DATE('31-12-2028','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (179, 'Body Lotion', 8, 1079, 'Vaseline', 320, 30, TO_DATE('31-08-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (180, 'Compact Powder', 8, 1080, 'Lakme', 550, 20, TO_DATE('30-06-2028','DD-MM-YYYY'));
1 row created.

```

```

INSERT INTO Product VALUES (181, 'Toy Car', 9, 1081, 'Hot Wheels', 600, 20, NULL);
1 row created.

INSERT INTO Product VALUES (182, 'Teddy Bear', 9, 1082, 'Funskool', 1200, 15, NULL);
1 row created.

INSERT INTO Product VALUES (183, 'Building Blocks', 9, 1083, 'Lego', 2500, 10, NULL);
1 row created.

INSERT INTO Product VALUES (184, 'Doll', 9, 1084, 'Barbie', 1500, 18, NULL);
1 row created.

INSERT INTO Product VALUES (185, 'Remote Car', 9, 1085, 'Maisto', 1800, 12, NULL);
1 row created.

INSERT INTO Product VALUES (186, 'Puzzle Game', 9, 1086, 'Funskool', 450, 25, NULL);
1 row created.

INSERT INTO Product VALUES (187, 'Board Game', 9, 1087, 'Hasbro', 1200, 15, NULL);
1 row created.

INSERT INTO Product VALUES (188, 'Toy Train', 9, 1088, 'Fisher Price', 2000, 8, NULL);
1 row created.

INSERT INTO Product VALUES (189, 'Rubik Cube', 9, 1089, 'Rubik', 500, 30, NULL);
1 row created.

INSERT INTO Product VALUES (190, 'Water Gun', 9, 1090, 'Nerf', 800, 20, NULL);
1 row created.

```

```

INSERT INTO Product VALUES (191, 'Notebook', 10, 1091, 'Classmate', 60, 100, NULL);
1 row created.

INSERT INTO Product VALUES (192, 'Pen Set', 10, 1092, 'Cello', 120, 50, NULL);
1 row created.

INSERT INTO Product VALUES (193, 'Pencil Box', 10, 1093, 'Camlin', 150, 40, NULL);
1 row created.

INSERT INTO Product VALUES (194, 'Eraser', 10, 1094, 'Apsara', 10, 100, NULL);
1 row created.

INSERT INTO Product VALUES (195, 'Sharpener', 10, 1095, 'Nataraj', 15, 80, NULL);
1 row created.

INSERT INTO Product VALUES (196, 'Marker Set', 10, 1096, 'Camlin', 180, 45, NULL);
1 row created.

INSERT INTO Product VALUES (197, 'Geometry Box', 10, 1097, 'Camlin', 250, 35, NULL);
1 row created.

INSERT INTO Product VALUES (198, 'File Folder', 10, 1098, 'Solo', 100, 60, NULL);
1 row created.

INSERT INTO Product VALUES (199, 'Sticky Notes', 10, 1099, 'Post-it', 80, 50, NULL);
1 row created.

INSERT INTO Product VALUES (200, 'A4 Paper', 10, 1100, 'JK Copier', 350, 30, NULL);
1 row created.

```

5. Data Retrieval

5.1 Display Category Records

```
SELECT * FROM Category;
```

CATEGORY_ID	CATEGORY_NAME	DESCRIPTION
1	Electronics	Electronic products
2	Clothing	Clothing products
3	Books	Educational books
4	Furniture	Home furniture
5	Groceries	Daily grocery items
6	Footwear	Shoes and sandals
7	Sports	Sports equipment
8	Beauty	Beauty products
9	Toys	Toys and games
10	Stationery	Stationery items

10 rows selected.

All category records are retrieved to verify successful insertion.

5.2 Display Product Records

All product records are retrieved to verify successful insertion.

```
Select * From Product;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
101 Laptop		1
1001 Dell		10
102 Smartphone		1
1002 Samsung		20
103 Tablet		1
1003 Lenovo		15

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
104 Keyboard		1
1004 HP		25
105 Mouse		1
1005 Logitech		30
106 Monitor		1
1006 LG		12

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
107	Printer			1
1007	Canon	12000	8	
108	Speaker			1
1008	JBL	3500	18	
109	Webcam			1
1009	Logitech	2500	20	
PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
110	Headphones			1
1010	Sony	4500	22	
111	T-Shirt			2
1011	Puma	1200	30	
112	Jeans			2
1012	Levis	2500	25	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
113	Formal Shirt			2
1013	Peter England	1800	20	
114	Kurti			2
1014	Biba	2200	18	
115	Saree			2
1015	Silk India	4500	12	
PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
116	Hoodie			2
1016	Adidas	2800	15	
117	Jacket			2
1017	Allen Solly	3500	10	
118	Shorts			2
1018	Nike	1500	22	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
119	Skirt			2
1019	Zara	2000	14	
120	Sweater			2
1020	Van Heusen	2400	16	
121	DBMS Book			3
1021	McGraw	650	15	
PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
122	Python Programming			3
1022	OReilly	850	12	
123	Java Programming			3
1023	Pearson	900	10	
124	Data Structures			3
1024	McGraw	750	14	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
125	Machine Learning			3
1025	Wiley	1100	8	
126	Artificial Intelligence			3
1026	Pearson	1250	9	
127	Web Development			3
1027	Packt	950	11	
PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
128	Computer Networks			3
1028	McGraw	800	13	
129	Operating Systems			3
1029	Pearson	700	16	
130	Software Engineering			3
1030	Wiley	850	10	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
131	Computer Table			4
1031	WoodCraft	4500	8	
132	Office Chair			4
1032	FeatherLite	6500	10	
133	Dining Table			4
1033	UrbanWood	12000	5	
PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
134	Sofa			4
1034	IKEA	25000	4	
135	Bookshelf			4
1035	HomeTown	7000	7	
136	Bed			4
1036	Wakefit	30000	3	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
137	Wardrobe			4
1037	Godrej	22000	6	
138	Study Table			4
1038	Nilkamal	5500	9	
139	Coffee Table			4
1039	UrbanLadder	6500	8	
PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
140	TV Stand			4
1040	HomeTown	8000	5	
141	Rice			5
1041	India Gate	800	30	30-JUN-27
142	Wheat Flour			5
1042	Aashirvaad	450	25	15-JUL-27

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
143	Cooking Oil			5
1043	Fortune	180	40	30-JUN-27
144	Sugar			5
1044	Madhur	55	50	31-DEC-27
145	Salt			5
1045	Tata	30	45	31-DEC-28
146	Toor Dal			5
1046	Tata Sampann	160	35	30-SEP-27
147	Biscuits			5
1047	Parle	40	60	15-MAY-27
148	Tea			5
1048	Tata Tea	250	30	31-AUG-27

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
149	Coffee			5
1049	Bru	300	25	30-NOV-27
150	Noodles			5
1050	Maggi	60	50	31-OCT-27
151	Running Shoes			6
1051	Nike	3500	18	
152	Sports Sandals			6
1052	Adidas	2200	15	
153	Formal Shoes			6
1053	Bata	2800	12	
154	Sneakers			6
1054	Puma	3200	20	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
155	Slippers			6
1055	Sparx	600	35	
156	Heels			6
1056	Metro	2500	14	
157	Flip Flops			6
1057	Hawaii	400	40	
158	Loafers			6
1058	Clarks	4500	10	
159	School Shoes			6
1059	Bata	1200	25	
160	Walking Shoes			6
1060	Reebok	2800	18	

PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
161	Football		7
1061	Nivia	900	25
162	Cricket Bat		7
1062	SS	2800	12
163	Cricket Ball		7
1063	SG	450	30
PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
164	Tennis Racket		7
1064	Yonex	3500	10
165	Badminton Racket		7
1065	Li-Ning	2500	15
166	Basketball		7
1066	Spalding	1800	14

PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
167	Volleyball		7
1067	Cosco	1200	18
168	Skipping Rope		7
1068	Nivia	300	35
169	Yoga Mat		7
1069	Boldfit	800	25
PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
170	Dumbbells		7
1070	Kobo	1500	12
171	Face Cream		8
1071	Nivea	450	25 31-DEC-27
172	Shampoo		8
1072	Dove	350	30 31-DEC-27

PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
173	Lipstick		8
1073	Lakme	599	20 30-JUN-28
174	Face Wash		8
1074	Himalaya	250	35 31-DEC-27
175	Moisturizer		8
1075	Ponds	400	28 30-SEP-27
PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
176	Sunscreen		8
1076	Neutrogena	650	22 31-DEC-27
177	Kajal		8
1077	Maybelline	300	25 30-JUN-28
178	Perfume		8
1078	Fogg	750	18 31-DEC-28

PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
179	Body Lotion		8
1079	Vaseline	320	30 31-AUG-27
180	Compact Powder		8
1080	Lakme	550	20 30-JUN-28
181	Toy Car		9
1081	Hot Wheels	600	20
182	Teddy Bear		9
1082	Funskool	1200	15
183	Building Blocks		9
1083	Lego	2500	10
184	Doll		9
1084	Barbie	1500	18

PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
185	Remote Car		9
1085	Maisto	1800	12
186	Puzzle Game		9
1086	Funskool	450	25
187	Board Game		9
1087	Hasbro	1200	15
188	Toy Train		9
1088	Fisher Price	2000	8
189	Rubik Cube		9
1089	Rubik	500	30
190	Water Gun		9
1090	Nerf	800	20

PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
191	Notebook		10
1091	Classmate	60	100
192	Pen Set		10
1092	Cello	120	50
193	Pencil Box		10
1093	Camlin	150	40
194	Eraser		10
1094	Apsara	10	100
195	Sharpener		10
1095	Nataraj	15	80
196	Marker Set		10
1096	Camlin	180	45

PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
197	Geometry Box		10
1097	Camlin	250	35
198	File Folder		10
1098	Solo	100	60
199	Sticky Notes		10
1099	Post-it	80	50
200	A4 Paper		10
1100	JK Copier	350	30

100 rows selected.

PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
191	Notebook		10
1091	Classmate	60	100
192	Pen Set		10
1092	Cello	120	50
193	Pencil Box		10
1093	Camlin	150	40
194	Eraser		10
1094	Apsara	10	100
195	Sharpener		10
1095	Nataraj	15	80
196	Marker Set		10
1096	Camlin	180	45

PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
197	Geometry Box		10
1097	Camlin	250	35
198	File Folder		10
1098	Solo	100	60
199	Sticky Notes		10
1099	Post-it	80	50
200	A4 Paper		10
1100	JK Copier	350	30

100 rows selected.

6. Product Update

Update the price of DBMS Book (Product_ID 121) and

```

Update Product
Set Price=700, Stock=30
Where Product_ID=121;
1 row updated.

UPDATE Product
SET Price = 7000,
    Stock = 12
WHERE Product_ID = 132;
1 row updated.

UPDATE Product
SET Price = 3000,
    Stock = 25
WHERE Product_ID = 160;
1 row updated.

SELECT *
FROM Product
WHERE Product_ID IN (121, 150, 132);

```

Verify the updated price by selecting the record.

PRODUCT_ID	PRODUCT_NAME		CATEGORY_ID
SUPPLIER_ID	BRAND	PRICE	STOCK EXPIRY_DA
121	DBMS Book		3
1021	McGraw	700	30
132	Office Chair		4
1032	FeatherLite	7000	12
160	Walking Shoes		6
1060	Reebok	3000	25

7. Product Deletion

Delete the product Shorts (Product_ID 118), Shampoo(Product_ID 172) and Pen Set (Product_ID 192) from the Product table.

```
DELETE FROM Product
WHERE Product_ID = 118;
1 row deleted.

DELETE FROM Product
WHERE Product_ID = 172;
1 row deleted.

DELETE FROM Product
WHERE Product_ID = 192;
1 row deleted.
```

Verify the deletion by displaying the remaining product records.

Select * From Product;

PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
101	Laptop	55000	10	1
1001	Dell			
102	Smartphone	25000	20	1
1002	Samsung			
103	Tablet	18000	15	1
1003	Lenovo			
PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
104	Keyboard	1200	25	1
1004	HP			
105	Mouse	800	30	1
1005	Logitech			
106	Monitor	15000	12	1
1006	LG			
PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
107	Printer	12000	8	1
1007	Canon			
108	Speaker	3500	18	1
1008	JBL			
109	Webcam	2500	20	1
1009	Logitech			

PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
110	Headphones	4500	22	1
1010	Sony			
111	T-Shirt	1200	30	2
1011	Puma			
112	Jeans	2500	25	2
1012	Levis			
PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
113	Formal Shirt	1800	20	2
1013	Peter England			
114	Kurti	2200	18	2
1014	Biba			
115	Saree	4500	12	2
1015	Silk India			
PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
116	Hoodie	2800	15	2
1016	Adidas			
117	Jacket	3500	10	2
1017	Allen Solly			
119	Skirt	2000	14	2
1019	Zara			

PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
120	Sweater	2400	16	2
1020	Van Heusen			
121	DBMS Book	700	30	3
1021	McGraw			
122	Python Programming	850	12	3
1022	O'Reilly			
PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
123	Java Programming	900	10	3
1023	Pearson			
124	Data Structures	750	14	3
1024	McGraw			
125	Machine Learning	1100	8	3
1025	Wiley			
PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
126	Artificial Intelligence	1250	9	3
1026	Pearson			
127	Web Development	950	11	3
1027	Packt			
128	Computer Networks	800	13	3
1028	McGraw			

PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
129	Operating Systems	700	16	3
1029	Pearson			
130	Software Engineering	850	10	3
1030	Wiley			
131	Computer Table	4500	8	4
1031	WoodCraft			
PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
132	Office Chair	7000	12	4
1032	FeatherLite			
133	Dining Table	12000	5	4
1033	UrbanWood			
134	Sofa	25000	4	4
1034	IKEA			
PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA
135	Bookshelf	7000	7	4
1035	HomeTown			
136	Bed	30000	3	4
1036	Nakefit			
137	Wardrobe	22000	6	4
1037	Godrej			

PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
-----		-----		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA

138	Study Table	5500	9	4
1038	Nilkamal			
139	Coffee Table	6500	8	4
1039	UrbanLadder			
140	TV Stand	8000	5	4
1040	HomeTown			
PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
-----		-----		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA

141	Rice	800	30	30-JUN-27
1041	India Gate			
142	Wheat Flour	450	25	15-JUL-27
1042	Aashirvaad			
143	Cooking Oil	180	40	30-JUN-27
1043	Fortune			
PRODUCT_ID PRODUCT_NAME		CATEGORY_ID		
-----		-----		
SUPPLIER_ID	BRAND	PRICE	STOCK	EXPIRY_DA

144	Sugar	55	50	31-DEC-27
1044	Madhur			
145	Salt	30	45	31-DEC-28
1045	Tata			
146	Toor Dal	160	35	30-SEP-27
1046	Tata Sampann			

PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
147	Biscuits	40	60	15-MAY-27	5
1847	Parle				
148	Tea	250	30	31-AUG-27	5
1848	Tata Tea				
149	Coffee	300	25	30-NOV-27	5
1849	Bru				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
150	Noodles	60	50	31-OCT-27	5
1850	Maggi				
151	Running Shoes	3500	18		6
1851	Nike				
152	Sports Sandals	2200	15		6
1852	Adidas				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
153	Formal Shoes	2800	12		6
1853	Bata				
154	Sneakers	3200	20		6
1854	Puma				
155	Slippers	600	35		6
1855	Sparx				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
156	Heels	2500	14		6
1856	Metro				
157	Flip Flops	400	40		6
1857	Hawaii				
158	Loafers	4500	10		6
1858	Clarks				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
159	School Shoes	1200	25		6
1859	Bata				
160	Walking Shoes	3000	25		6
1860	Reebok				
161	Football	900	25		7
1861	Nivia				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
162	Cricket Bat	2800	12		7
1862	SS				
163	Cricket Ball	450	30		7
1863	SG				
164	Tennis Racket	3500	10		7
1864	Yonex				

PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
165	Badminton Racket	2500	15		7
1865	Li-Ning				
166	Basketball	1800	14		7
1866	Spalding				
167	Volleyball	1200	18		7
1867	Cosco				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
168	Skipping Rope	300	35		7
1868	Nivia				
169	Yoga Mat	800	25		7
1869	Boldfit				
170	Dumbbells	1500	12		7
1870	Kobo				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
171	Face Cream	450	25	31-DEC-27	8
1871	Nivea				
173	Lipstick	599	20	30-JUN-28	8
1873	Lakme				
174	Face Wash	250	35	31-DEC-27	8
1874	Himalaya				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
175	Moisturizer	400	28	30-SEP-27	8
1875	Ponds				
176	Sunscreen	650	22	31-DEC-27	8
1876	Neutrogena				
177	Kajal	300	25	30-JUN-28	8
1877	Maybelline				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
178	Perfume	750	18	31-DEC-28	8
1878	Fogg				
179	Body Lotion	320	30	31-AUG-27	8
1879	Vaseline				
180	Compact Powder	550	20	30-JUN-28	8
1880	Lakme				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
181	Toy Car	600	20		9
1881	Hot Wheels				
182	Teddy Bear	1200	15		9
1882	Funskool				
183	Building Blocks	2500	10		9
1883	Legs				

PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
184	Doll	1500	18		9
1884	Barbie				
185	Remote Car	1800	12		9
1885	Maisto				
186	Puzzle Game	450	25		9
1886	Funskool				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
187	Board Game	1200	15		9
1887	Hasbro				
188	Toy Train	2000	8		9
1888	Fisher Price				
189	Rubik Cube	500	30		9
1889	Rubik				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
190	Water Gun	800	20		9
1890	Nerf				
191	Notebook	60	100		10
1891	Classmate				
193	Pencil Box	150	40		10
1893	Camlin				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
194	Eraser	10	100		10
1894	Apsara				
195	Sharpener	15	80		10
1895	Nataraj				
196	Marker Set	180	45		10
1896	Camlin				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
197	Geometry Box	250	35		10
1897	Camlin				
198	File Folder	100	60		10
1898	Solo				
199	Sticky Notes	80	50		10
1899	Post-it				
PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	EXPIRY_DA	CATEGORY_ID
SUPPLIER_ID	BRAND				
200	A4 Paper	350	30		10
1100	JK Copier				

97 rows selected.

8. Category-wise Product Reports

8.1 Product Listing by Category

This report displays the category name along with the products belonging to each category, using an inner JOIN between Category and Product.

Select c.Category_ID,c.Category_Name,p.Product_ID,p.Product_Name,p.Price,p.Stock
from Category c join Product p
on c.Category_ID=p.Category_ID
Order By c.Category_ID,p.Category_ID;

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK
1	Electronics	101	Laptop	55000	10
		102	Smartphone	25000	20
		110	Headphones	4500	22
		109	Webcam	2500	20
1	Electronics	108	Speaker	3500	18
		107	Printer	12000	8
		106	Monitor	15000	12
1	Electronics	105	Mouse	800	30

2	Clothing	119	Skirt	2000	14
		117	Jacket	3500	10
		116	Hoodie	2800	15
		115	Saree	4500	12
2	Clothing	120	Sweater	2400	16
		113	Formal Shirt	1800	20
		112	Jeans	2500	25
2	Clothing	111	T-Shirt	1200	30

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID	PRICE	STOCK
2	Clothing	114	2200	18
3	Books	121	700	30
3	Books	130	850	10
3	Books	129	700	16
3	Books	128	800	13
3	Books	127	950	11
3	Books	126	1250	9
3	Books	125	1100	8
3	Books	124	750	14
3	Books	123	900	10
3	Books	122	850	12

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID	PRICE	STOCK
4	Furniture	139	6500	8
4	Furniture	138	5500	9
4	Furniture	137	22000	6
4	Furniture	136	30000	3
4	Furniture	140	8000	5
4	Furniture	134	25000	4
4	Furniture	133	12000	5
4	Furniture	132	7000	12
4	Furniture	131	4500	8
4	Furniture	135	7000	7

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID	PRICE	STOCK
5	Groceries	141	800	30
5	Groceries	142	450	25
5	Groceries	143	180	40
5	Groceries	144	55	50
5	Groceries	145	30	45
5	Groceries	146	160	35
5	Groceries	147	40	60
5	Groceries	148	250	30
5	Groceries	149	300	25
6	Footwear	150	60	50
6	Footwear	153	2800	12
6	Footwear	152	2200	15
6	Footwear	160	3000	25
6	Footwear	159	1200	25
6	Footwear	158	4500	10
6	Footwear	157	400	40
6	Footwear	156	2500	14
6	Footwear	155	600	35

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Sneakers	6 Footwear	154	3200	20	
Running Shoes	6 Footwear	151	3500	18	
Football	7 Sports	161	900	25	
CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Yoga Mat	7 Sports	169	800	25	
Skipping Rope	7 Sports	168	300	35	
Volleyball	7 Sports	167	1200	18	
CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Dumbbells	7 Sports	170	1500	12	
Badminton Racket	7 Sports	165	2500	15	
Tennis Racket	7 Sports	164	3500	10	

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Cricket Ball	7 Sports	163	450	30	
Cricket Bat	7 Sports	162	2800	12	
Basketball	7 Sports	166	1800	14	
CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Face Wash	8 Beauty	174	250	35	
Lipstick	8 Beauty	173	599	20	
Face Cream	8 Beauty	171	450	25	
CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Moisturizer	8 Beauty	175	400	28	
Compact Powder	8 Beauty	180	550	20	
Body Lotion	8 Beauty	179	320	30	

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Perfume	8 Beauty	178	750	18	
Kajal	8 Beauty	177	300	25	
Sunscreen	8 Beauty	176	650	22	
CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Toy Car	9 Toys	181	600	20	
Teddy Bear	9 Toys	182	1200	15	
Water Gun	9 Toys	190	800	20	
CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Rubik Cube	9 Toys	189	500	30	
Toy Train	9 Toys	188	2000	8	
Board Game	9 Toys	187	1200	15	

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Puzzle Game	9 Toys	186	450	25	
Remote Car	9 Toys	185	1800	12	
Doll	9 Toys	184	1500	18	
CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Building Blocks	9 Toys	183	2500	10	
Sharpener	10 Stationery	195	15	80	
Marker Set	10 Stationery	196	180	45	
CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Geometry Box	10 Stationery	197	250	35	
File Folder	10 Stationery	198	100	60	
A4 Paper	10 Stationery	200	350	30	

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Eraser	10 Stationery	194	10	100	
Pencil Box	10 Stationery	193	150	40	
Notebook	10 Stationery	191	60	100	
CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID			
PRODUCT_NAME			PRICE	STOCK	
Sticky Notes	10 Stationery	199	80	50	
97 rows selected.					

Conclusion

The Category and Product tables were successfully designed, created, and populated with sample data. CRUD operations (Insert, Select, Update, Delete) were performed and verified, and a JOIN query was used to generate a category-wise product report, demonstrating a working one-to-many relationship between Category and Product.